

# Circular Economy Ontology Network (CEON) - Resource ODP

## Metadata

### IRI

<http://w3id.org/CEON/ontology/resourceODP/>

### Title

Circular Economy Ontology Network (CEON) - Resource ODP

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### Version Iri

<http://w3id.org/CEON/ontology/resourceODP/0.2/>

### Version Info

0.2

### Preferred Namespace Prefix

resourceODP

### Preferred Namespace Uri

<http://w3id.org/CEON/ontology/resourceODP/>

### Description

A core ODP of the CEON ontology network defining aspects of the resource concept.

### Covers Requirements

Covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-1,3, CVN-Composition-1,2, CVN-ResrouceType-4, C7-3, E2-2, E4-6, E5-1, E6-3, T3-1.

## Classes

| Batch of objects <sup>c</sup> |                                                                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IRI                           | <a href="http://w3id.org/CEON/ontology/resourceODP/BatchOf0bjects">http://w3id.org/CEON/ontology/resourceODP/BatchOf0bjects</a>                                                                                                                      |
| Description                   | A batch of objects is a collection of physical objects that are of the same type, e.g. a set of items (product objects) adhering to the same product model.                                                                                          |
| Sub Class Of                  | <a href="#">Resource</a> <sup>c</sup><br><a href="#">has physical object</a> <sup>op</sup> some <a href="#">Physical object</a> <sup>c</sup> and <a href="#">has physical object</a> <sup>op</sup> only <a href="#">Physical object</a> <sup>c</sup> |
| In Domain Of                  | <a href="#">batch size</a> <sup>dp</sup>                                                                                                                                                                                                             |
| In Range Of                   | <a href="#">has batch</a> <sup>op</sup>                                                                                                                                                                                                              |
| Restriction                   | <a href="#">batch size</a> <sup>dp</sup> <i>exactly</i> 1 <a href="#">Batch of objects</a> <sup>c</sup>                                                                                                                                              |

## Composition<sup>C</sup>

|                    |                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resourceODP/Composition">http://w3id.org/CE0N/ontology/resourceODP/Composition</a>                                                                                                  |
| <b>Description</b> | A composition is used to represent how a whole or mixture is made up. E.g., for chemical elements composing a chemical structure, the composition should hold information of the ratio of the composing chemical elements. |

## Constituent<sup>C</sup>

|                    |                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resourceODP/Constituent">http://w3id.org/CE0N/ontology/resourceODP/Constituent</a> |
| <b>Description</b> | A constituent is a component of object.                                                                                   |
| <b>In Range Of</b> | <a href="#">has constituent</a> <sup>op</sup>                                                                             |

## Digital object<sup>C</sup>

|                     |                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resourceODP/DigitalObject">http://w3id.org/CE0N/ontology/resourceODP/DigitalObject</a> |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup>                                                                                         |

## Information<sup>C</sup>

|                     |                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resourceODP/Information">http://w3id.org/CE0N/ontology/resourceODP/Information</a>                                                                      |
| <b>Description</b>  | Information is an abstract concept that represents any kind of interpretations. For instance, information can be data generated by software systems or data used by people for communications. |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup>                                                                                                                                                          |
| <b>In Domain Of</b> | <a href="#">containsInformation</a> <sup>op</sup><br><a href="#">isAbout</a> <sup>op</sup>                                                                                                     |
| <b>In Range Of</b>  | <a href="#">containsInformation</a> <sup>op</sup><br><a href="#">is realization of</a> <sup>op</sup>                                                                                           |

## Matter<sup>C</sup>

|                    |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resourceODP/Matter">http://w3id.org/CE0N/ontology/resourceODP/Matter</a> |
| <b>Description</b> | A matter is a physical substance.                                                                               |
| <b>In Range Of</b> | <a href="#">has matter</a> <sup>op</sup>                                                                        |

## Physical object<sup>C</sup>

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resource0DP/PhysicalObject">http://w3id.org/CE0N/ontology/resource0DP/PhysicalObject</a>                                                                                                                                                                                                                                                                                               |
| <b>Description</b>  | A physical object is a collection of matter.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup><br><a href="#">has constituent</a> <sup>op</sup> some <a href="#">Constituent</a> <sup>C</sup> <i>and</i> <a href="#">has constituent</a> <sup>op</sup> only <a href="#">Constituent</a> <sup>C</sup><br><a href="#">has matter</a> <sup>op</sup> only <a href="#">Matter</a> <sup>C</sup> <i>and</i> <a href="#">has matter</a> <sup>op</sup> some <a href="#">Matter</a> <sup>C</sup> |
| <b>In Domain Of</b> | <a href="#">has constituent</a> <sup>op</sup><br><a href="#">has matter</a> <sup>op</sup>                                                                                                                                                                                                                                                                                                                                     |
| <b>In Range Of</b>  | <a href="#">has physical object</a> <sup>op</sup>                                                                                                                                                                                                                                                                                                                                                                             |

## Resource<sup>C</sup>

|                       |                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>            | <a href="http://w3id.org/CE0N/ontology/resource0DP/Resource">http://w3id.org/CE0N/ontology/resource0DP/Resource</a>                                                                                                                     |
| <b>Description</b>    | A resource able to be handled in the context of a circular value network, e.g. data generated by software systems in the CVN, materials or prooducts as physical objects handled in the CVN.                                            |
| <b>In Domain Of</b>   | <a href="#">hasResourceLocation</a> <sup>op</sup>                                                                                                                                                                                       |
| <b>Super Class Of</b> | <a href="#">Batch of objects</a> <sup>C</sup><br><a href="#">Digital object</a> <sup>C</sup><br><a href="#">Information</a> <sup>C</sup><br><a href="#">Physical object</a> <sup>C</sup><br><a href="#">Set of objects</a> <sup>C</sup> |

## Set of objects<sup>C</sup>

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resource0DP/SetOfObjects">http://w3id.org/CE0N/ontology/resource0DP/SetOfObjects</a>                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b>  | A set of objects is a set of physical objects (items) that can be of different types, i.e. different kinds of items.                                                                                                                                                                                                                                                                                                                                            |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup><br><a href="#">has batch</a> <sup>op</sup> some <a href="#">Batch of objects</a> <sup>C</sup> <i>and</i> <a href="#">has batch</a> <sup>op</sup> only <a href="#">Batch of objects</a> <sup>C</sup><br><a href="#">has physical object</a> <sup>op</sup> only <a href="#">Physical object</a> <sup>C</sup> <i>and</i> <a href="#">has physical object</a> <sup>op</sup> some <a href="#">Physical object</a> <sup>C</sup> |

## Geometry<sup>c</sup>

|                     |                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://www.opengis.net/ont/geosparql#Geometry">http://www.opengis.net/ont/geosparql#Geometry</a>                |
| <b>Description</b>  | A coherent set of direct positions in space. The positions are held within a Spatial Reference System (SRS).             |
| <b>In Domain Of</b> | <a href="#">as GML</a> <sup>dp</sup><br><a href="#">as GeoJSON</a> <sup>dp</sup><br><a href="#">as WKT</a> <sup>dp</sup> |
| <b>In Range Of</b>  | <a href="#">hasResourceLocation</a> <sup>op</sup>                                                                        |

## Object Properties

### contains information<sup>op</sup>

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/resourceODP/containsInformation">http://w3id.org/CE0N/ontology/resourceODP/containsInformation</a> |
| <b>Sub Property Of</b> | <a href="#">hasPart</a> <sup>op</sup>                                                                                                     |
| <b>Domain</b>          | <a href="#">Information</a> <sup>c</sup>                                                                                                  |
| <b>Range</b>           | <a href="#">Information</a> <sup>c</sup>                                                                                                  |

### has batch<sup>op</sup>

|                    |                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resourceODP/hasBatch">http://w3id.org/CE0N/ontology/resourceODP/hasBatch</a>                                         |
| <b>Description</b> | hasBatch intends to represent that a set or batch of objects can be captured by a number of batches where each batch contains a number of physical objects. |
| <b>Domain</b>      | <a href="#">Batch of objects</a> <sup>c</sup> <i>or</i> <a href="#">Set of objects</a> <sup>c</sup>                                                         |
| <b>Range</b>       | <a href="#">Batch of objects</a> <sup>c</sup>                                                                                                               |

### has constituent<sup>op</sup>

|                        |                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/resourceODP/hasConstituent">http://w3id.org/CE0N/ontology/resourceODP/hasConstituent</a> |
| <b>Description</b>     | hasConstituent intends to represent that a physical object can have a collection of composing components.                       |
| <b>Sub Property Of</b> | <a href="#">hasPart</a> <sup>op</sup>                                                                                           |
| <b>Domain</b>          | <a href="#">Physical object</a> <sup>c</sup>                                                                                    |
| <b>Range</b>           | <a href="#">Constituent</a> <sup>c</sup>                                                                                        |

## has matter<sup>op</sup>

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasMatter">http://w3id.org/CE0N/ontology/resource0DP/hasMatter</a> |
| <b>Description</b> | hasMatter intends to represent that a physical object can have a collection of matter.                                |
| <b>Domain</b>      | <a href="#">Physical object</a> <sup>c</sup>                                                                          |
| <b>Range</b>       | <a href="#">Matter</a> <sup>c</sup>                                                                                   |

## has part<sup>op</sup>

|                          |                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>               | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasPart">http://w3id.org/CE0N/ontology/resource0DP/hasPart</a>                                         |
| <b>Super Property Of</b> | <ul style="list-style-type: none"><li>• <a href="#">containsInformation</a><sup>op</sup></li><li>• <a href="#">has constituent</a><sup>op</sup></li></ul> |

## has physical object<sup>op</sup>

|                    |                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject">http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject</a> |
| <b>Description</b> | hasPhysicalObject intends to represent that a batch of objects or a set of objects can have composing components of physical objects. |
| <b>Domain</b>      | <a href="#">Set of objects</a> <sup>c</sup> or <a href="#">Batch of objects</a> <sup>c</sup>                                          |
| <b>Range</b>       | <a href="#">Physical object</a> <sup>c</sup>                                                                                          |

## has resource location<sup>op</sup>

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation">http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation</a> |
| <b>Domain</b> | <a href="#">Resource</a> <sup>c</sup>                                                                                                     |
| <b>Range</b>  | <a href="#">Geometry</a> <sup>c</sup>                                                                                                     |

## is about<sup>op</sup>

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/isAbout">http://w3id.org/CE0N/ontology/resource0DP/isAbout</a> |
| <b>Domain</b> | <a href="#">Information</a> <sup>c</sup>                                                                          |

## is realization of<sup>op</sup>

|              |                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>   | <a href="http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf">http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf</a> |
| <b>Range</b> | <a href="#">Information</a> <sup>c</sup>                                                                                          |

## Datatype Properties

batch size<sup>dp</sup>

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resource0DP/batchSize">http://w3id.org/CE0N/ontology/resource0DP/batchSize</a> |
| <b>Description</b> | batchSize intends to represent how many physical objects are belong to a batch of objects.                            |
| <b>Domain</b>      | <a href="#">Batch of objects</a> <sup>c</sup>                                                                         |
| <b>Range</b>       | <a href="#">xsd:nonNegativeInteger</a>                                                                                |

as GML<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asGML">http://www.opengis.net/ont/geosparql#asGML</a>                                                                                             |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The GML serialization of a Geometry.                                                                                                                                                            |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:gmlLiteral</a>                                                                                                                                                                  |

as GeoJSON<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asGeoJSON">http://www.opengis.net/ont/geosparql#asGeoJSON</a>                                                                                     |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The GeoJSON serialization of a Geometry.                                                                                                                                                        |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:geoJSONLiteral</a>                                                                                                                                                              |

as WKT<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asWKT">http://www.opengis.net/ont/geosparql#asWKT</a>                                                                                             |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The WKT serialization of a Geometry.                                                                                                                                                            |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:wktLiteral</a>                                                                                                                                                                  |

## Annotation Properties

description<sup>ap</sup>

IRI <http://purl.org/dc/elements/1.1/description>

contributor<sup>ap</sup>

IRI <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

IRI <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

IRI <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

IRI <http://purl.org/dc/terms/description>

license<sup>ap</sup>

IRI <http://purl.org/dc/terms/license>

title<sup>ap</sup>

IRI <http://purl.org/dc/terms/title>

preferred namespace prefix<sup>ap</sup>

IRI <http://purl.org/vocab/vann/preferredNamespacePrefix>

preferred namespace uri<sup>ap</sup>

IRI <http://purl.org/vocab/vann/preferredNamespaceUri>

covers requirements<sup>ap</sup>

IRI <http://www.ontologydesignpatterns.org/schemas/cpannotation/schema.owl#coversRequirements>

definition<sup>ap</sup>

IRI <http://www.w3.org/2004/02/skos/core#definition>

pref label<sup>ap</sup>

IRI <http://www.w3.org/2004/02/skos/core#prefLabel>

## Namespaces

|         |                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| :       | <a href="http://w3id.org/CEON/ontology/resourceODP/">http://w3id.org/CEON/ontology/resourceODP/</a>                                                       |
| dc      | <a href="http://purl.org/dc/elements/1.1/">http://purl.org/dc/elements/1.1/</a>                                                                           |
| dcterms | <a href="http://purl.org/dc/terms/">http://purl.org/dc/terms/</a>                                                                                         |
| geo     | <a href="http://www.opengis.net/ont/geosparql#">http://www.opengis.net/ont/geosparql#</a>                                                                 |
| odp     | <a href="http://www.ontologydesignpatterns.org/schemas/cpannotationschema.owl#">http://www.ontologydesignpatterns.org/schemas/cpannotationschema.owl#</a> |
| owl     | <a href="http://www.w3.org/2002/07/owl#">http://www.w3.org/2002/07/owl#</a>                                                                               |
| prov    | <a href="http://www.w3.org/ns/prov#">http://www.w3.org/ns/prov#</a>                                                                                       |
| rdf     | <a href="http://www.w3.org/1999/02/22-rdf-syntax-ns#">http://www.w3.org/1999/02/22-rdf-syntax-ns#</a>                                                     |
| rdfs    | <a href="http://www.w3.org/2000/01/rdf-schema#">http://www.w3.org/2000/01/rdf-schema#</a>                                                                 |
| skos    | <a href="http://www.w3.org/2004/02/skos/core#">http://www.w3.org/2004/02/skos/core#</a>                                                                   |
| vann    | <a href="http://purl.org/vocab/vann/">http://purl.org/vocab/vann/</a>                                                                                     |
| xsd     | <a href="http://www.w3.org/2001/XMLSchema#">http://www.w3.org/2001/XMLSchema#</a>                                                                         |

## Legend

|    |                       |
|----|-----------------------|
| c  | Classes               |
| op | Object Properties     |
| dp | Datatype Properties   |
| ap | Annotation Properties |